Chapter 13

Late Blight of Tomato

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Abstract

Late blight (LB), caused by the oomycete *Phytophthora infestans* (Mont.) de Bary, is one of the most destructive diseases of tomato (*Solanum lycopersicum* L.) and potato (*S. tuberosum* L.) worldwide, causing significant economic losses annually. The success of *P. infestans* as a pathogen originates from its effective asexual and sexual life cycles, as well as its remarkable capacity to rapidly overcome plant resistance genes, a result of its high evolutionary potential. The most sustainable strategy to manage tomato LB would be to deploy an integrated system including cultural practices, fungicide application, and the use of cultivars with broad-spectrum genetic resistance against LB. Prior to the reemergence of LB in the late 1980s, cultural practices in combination with fungicide applications were highly effective measures to control the tomato LB. However, with the appearance of new and more aggressive isolates of *P. infestans*, many of which are resistant to LB-specific systemic fungicides, the greatest contribution to tomato LB control in the future will have to be through the development of cultivars with improved genetic resistance. Thus far, a number of major LB-resistance genes and quantitative trait loci (QTLs) have been identified in tomato and several breeding lines and cultivars, with improved resistance developed. Research is also underway to identify additional resistance genes or QTLs and to pyramid multiple resistance factors in order to develop stronger and more durable resistance. Further, as exemplified by the fast progress in potato LB research and conservation of LB signaling pathways between potato and tomato, detailed knowledge of the pathogen effectors in combination with high-throughput genomics technology will facilitate a better understanding of the LB disease and host-pathogen interactions, which in turn may lead to development of tomatoes with more durable resistance.

Introduction

Tomato, *Solanum lycopersicum* L. (formerly known as *Lycopersicon esculentum* Mill.) is a major vegetable crop worldwide (FAOSTAT 2011). Tomato is thought to have originated in the Andean region of South America, now encompassed by parts of Peru, Chile, Colombia, Ecuador, and Bolivia (Jenkins 1948; Rick 1978), and domesticated in Mexico (Rick 1976). Although a tropical plant, it is grown in almost every corner of the world (see below). The broad

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spectrum use of tomato, including its importance in a healthy and balanced diet, has resulted in increased production of both processed and fresh varieties worldwide in the recent past.

Disease is the number one concern to both fresh-market and processing tomato industries throughout the world, and economic losses resulting from crop damage or disease control measures are significant (http://faostat.fao.org/). More than 200 pests and diseases have been identified as tomato pathogens hampering its production (Lukyanenko 1991). Among these, many are frequently occurring diseases caused by fungi, oomycetes, bacteria, viruses, and nematodes (listed in Foolad 2007). Late blight (LB), caused by the oomycete *Phytophthora infestans* (Mont.) de Bary, is one of the most destructive diseases of tomato as well as potato (*Solanum tuberosum* L.) worldwide, causing significant economic losses annually (reviewed in Foolad et al. 2008 and Nowicki et al. 2012). The pathogen is best known for its role in the Irish potato famine, where it caused the loss of more than a million lives (Andrivon 1996).

When left uncontrolled, *P. infestans* can destroy a tomato or potato crop within several days. The success of *P. infestans* as a pathogen originates from its effective asexual and sexual life cycles as well as its remarkable capacity to rapidly overcome plant resistance genes (Foolad et al. 2008; Nowicki et al. 2012). The latter feature has led researchers to describe *P. infestans* as a pathogen with a "high evolutionary potential" (Raffaele et al. 2010b). Evolutionary and comparative analyses of the *P. infestans* genome reveal the peculiar architecture that underpins the pathogen's accelerated adaptation to host plants (Raffaele et al. 2010a; Vleeshouwers et al. 2011). The ability of *P. infestans* to propagate asexually and through sexual mating results in rapid reproduction, fast moving epidemics, and increased genetic diversity and survival (Fry 2008). A sustainable control of the LB disease requires integration of cultural practices, fungicide applications, and use of resistant cultivars (Fry and Goodwin 1997; Nowicki et al. 2012).

At the first stage of infection, *P. infestans* penetrates the plant and translocates effector proteins inside the host cells. Specific effectors can act as avirulence (*Avr*) factors and activate corresponding host-plant resistance genes (*R*-genes) according to the gene-for-gene resistance model (Hardham and Blackman 2010; Vleeshouwers et al. 2011; Nowicki et al. 2012). Upon recognition of the effector by the host-plant resistance protein (R-protein), effector-triggered immunity (ETI) is activated, often resulting in the hypersensitive response (HR). Thus far all cloned *R*-genes against *P. infestans* are of potato origin (Jia et al. 2010; Vleeshouwers et al. 2011), although a conservation of the LB-signaling pathways has been proven for both potato and tomato (Jia et al. 2009; Faino et al. 2010; Jia et al. 2010). Despite the current lack of cloned tomato *R*-genes, several sources of LB-resistance genes have been identified, and a few have been mapped to tomato chromosomes and successfully introduced into several tomato breeding lines and hybrid cultivars via plant breeding (Foolad et al. 2008; Gardner and Panthee 2010a; Gardner and Panthee 2010b; Panthee and Gardner 2010; Merk and Foolad 2011, Merk et al. 2012). Considering the resurgence of LB disease in the recent past, along with the decreasing effectiveness of fungicide treatments, it is prudent to conduct further research to identify and characterize new sources of resistance and develop new resistant cultivars by pyramiding multiple resistance genes. Currently, several research groups around the world are working towards this goal. In this chapter we summarize and discuss the current understanding of *P. infestans*, its effects on tomato production, and the genetics and breeding of LB resistance in tomato.

Significance of Tomato

Origin of the Species and Crop Production

The cultivated tomato (*S. lycopersicum* L.) originated from the Andean region, part of modern-day Chile, Boliva, Ecuador, Colombia, and Peru.

The exact time and place of tomato domestication is not clearly known; however, by the fifteenth century the crop had reached a fairly advanced stage of domestication. Tomato was exported to Europe during that time, and subsequent further intensification of the process occurred. Since the twentieth century, human beings have created a huge array of morphologically different cultivars and forms from the single species *S. lycopersicum* via plant breeding (Foolad et al. 2008). Worldwide efforts aimed at domestication, research, and breeding activities have resulted in modern tomato varieties (mostly hybrids) developed in all shapes, colors, and sizes. Modern tomato genomics knowledge has transformed breeding from an individually based activity to multidisciplinary teamwork, focused on exploiting genes from a tremendous tomato germplasm, for high efficiency breeding (Bai and Lindhout 2007).

Improvement in tomato cultivars is necessitated by increased tomato production in response to global demand for this commodity. In 2009, the total net economic value of tomato was ~Int.$56 billion, which ranked it fourth among all crop species (after rice, wheat, and soybeans) and first among all vegetable crops (FAOSTAT 2011). Although a tropical plant, tomato is grown in almost every region of the world, from the tropics to within a few degrees of the Arctic Circle. When outdoor production is restricted because of cold temperatures, the tomato is grown in greenhouses. Major tomato-producing countries (in descending order of tonnage as of 2009) include China, the United States, India, Turkey, and Egypt, followed by Italy, Iran, Spain, Brazil, Mexico, Russian Federation, and Uzbekistan (FAOSTAT 2011). In North America, production takes place in the U.S., Canada, and Mexico, comprising a total of approximately 310,000 ha.

Tomatoes are an important part of a diverse and balanced diet. In addition to tomatoes that are consumed as raw vegetables or added to other food items, a variety of processed products such as pastes, juices, sauces, and soups are mass-produced and marketed. Although tomato does not rank high in nutritional value, by virtue of volume consumed, it contributes significantly to the dietary intake of vitamins A and C, essential minerals, and other nutrients. For example, tomato ranks first among all fruits and vegetables as a source of vitamins, minerals, and phenolic antioxidants in the U.S. diet (Rick 1980; Vinson et al. 1998). In addition, fresh and processed tomatoes are the richest sources of the antioxidant lycopene, which arguably protects cells from oxidants that have been linked to cancer (Ashrafi et al. 2011). Market demands require high global production output of tomato. In order to provide growers and consumers with high quality tomatoes, breeders have to overcome a number of issues threatening tomato production.

Low Genetic Diversity within the Cultivated Species

Crop plant genomes have evolved under human selection, which often has led to loss-of-function mutations, such as loss of seed dispersal through shattering in grains, loss of seed dormancy, and loss of long branches (Sim et al. 2009). At the same time, some agriculturally desirable characteristics result from gain-of-function mutations. Examples include disease and pest resistance, high fruit nutritional quality, and large size of fruit and seed (Foolad 2007; Sim et al. 2009). During and following its domestication, the cultivated tomato has undergone intensive selection and a few genetic bottlenecks occurred, resulting in narrow genetic variation within the cultigen. For example, tomatoes that were originally introduced to Europe by Spanish explorers furnished the entire genetic base for the modern tomato cultivars throughout the world (Foolad 2007). The low genetic diversity in the cultivated tomato is reflected by a low level of isozymes and DNA markers polymorphism across tomato breeding lines and commercial cultivars (Labate and Baldo 2005). It is estimated that only about 5% of the total genetic variation within Solanum section Lycopersicon (i.e.,

all tomato species) can be found within the cultivated species, *S. lycopersicum*, and that genes for many desirable agricultural characteristics do not exist in this species (Miller and Tanksley 1990; Foolad 2007). The related wild tomato species, however, are a rich source of desirable genes and characteristics for crop improvement, though they remain largely under-exploited. For instance, a panel of 31 *S. lycopersicum* lines was approximately as polymorphic as one population of *S. pimpinellifolium*, the closest wild relative of the cultivated tomato (Labate et al. 2009). The species with the greatest variability are *S. chilense*, *S. haborchaites*, *S. peruvianum,* and *S. pennellii*, whereas the least variable species are *S. cheesmanii* and *S. pimpinellifolium* (Foolad 2007). Recent advancements in molecular markers and MAS (marker-assisted selection) technology are expected to make tomato improvement via introgression from wild species more feasible. Further, recent investigations based on single nucleotide polymorphism (SNP) suggest the existence of several tomato genome regions with much higher diversity relative to other tested regions (Labate and Baldo 2005; Sim et al. 2009; Hamilton et al. 2012; Sim et al. 2012). These regions are associated with introgressions from wild relatives and their identification will be useful for both tomato improvement and germplasm conservation. Moreover, the recent identification of SNPs within the tomato cultigen (Sim et al. 2009; Hamilton et al. 2012; Sim et al. 2012) facilitates effective use of marker technology in breeding programs that mainly exploit elite tomato germplasm.

Significance of Late Blight

Historical Significance of the Disease

Late blight (LB) has been identified as a major disease of tomato and potato and is one of the most devastating plant diseases of all time. An unprotected tomato field can suffer yield losses reaching up to 100% because of LB infection (Nowicki et al. 2012). *Phytophthora infestans* – literally, "plant destroyer," in Greek – has been traced back to the same origin as tomatoes and potatoes, that is, the Andean region (Foolad et al. 2008; Vleeshouwers et al. 2011). Common origin for both the host and pathogen populations, initially suggested in the nineteenth century shortly after the Irish potato famine (de Bary 1876), has been recently confirmed by isozyme and DNA studies as well as pathogenicity similarities among Peruvian, U.S., and European isolates of *P. infestans* (reviewed in Foolad et al. 2008; Vleeshouwers et al. 2011; and Nowicki et al. 2012). Briefly, the pathogen incited the first recorded instance of potato LB in Philadelphia and New York City, in the U.S., in 1843. Because of weather patterns, winds spread the dehiscent pathogen sporangia to neighboring states, dispersing them, and thus increasing the area affected by LB. By 1845, LB was affecting crops from Illinois to Nova Scotia and from Virginia to Ontario. The disease then crossed the Atlantic Ocean from the U.S. to Europe in 1845 with a shipment of infested seed potatoes. Once *P. infestans* reached Ireland, a country that was strongly dependent on potatoes as a main source of food and predisposed to adverse political, social, and economic factors, widespread potato LB resulted in a near-complete destruction of the crop. This led to the death of one million people and the displacement of another million refugees, many of whom immigrated to the United States. Continued spread of the pathogen in subsequent years resulted in a worldwide distribution of LB by the beginning of the twentieth century and global devastation of potato and tomato crops.

Recent observations confirm virulence of the pathogen towards potato, but also tomato: Once an unprotected crop (field, greenhouse, and/or plastic-cover cultures) is infected by *P. infestans*, the whole crop can be devastated within seven to ten days (Foolad et al. 2008). Economic losses may be in the form of reduced yield, lower quality of the fruit (such as low specific gravity), diminished storability, and increased cost

associated with fungicide applications (Nowicki et al. 2012).

Disease Cycle and Development

The severe fiscal and social impact of the pathogen increased scientific interest in LB research. Discoveries in the disease biology from potato-driven research have also benefited understanding of tomato LB. *P. infestans* can be particularly destructive in areas where both tomatoes and potatoes are grown year-round, for example in the highland tropics of Africa, the Americas, Asia, and Europe (Nowicki et al. 2012). The unhampered success of *P. infestans* as a pathogen originates from its effective asexual and sexual reproduction. The asexual form serves as the major vehicle driving epidemics during the season. In this form, *P. infestans* produces thousands of sporangia per lesion on sporangiophores (Figure 13.1C, D). Sporangiophores are the indeterminate structures that aid in air dispersal of the sporangia through wind, rain, or wind-blown rain transportation. The disease cycle begins when sporangia land on host plant tissue, which must be covered with a film of water, which is necessary for the motile, germinated spore movement towards a penetration site (Hardham and Blackman 2010). Sporangia germination occurs either through the direct extension of germ tubes or by zoosporogenesis. The latter is stimulated by cool and moist conditions, and it is important in that it extends the range of weather conditions in which infections can occur. Direct germination of sporangium on host tissue occurs at temperatures above 21° C (optimally at 25° C) in a process taking between 8 and 48 hours. The sporangium can germinate directly at temperatures above 15° C and quickly develops mycelial growth. At temperatures below 21° C, up to eight biflagellate zoospores are released from the sporangia, with optimal zoospore formation occurring at 12° C. Motile zoospores penetrate through the film of water, detach their flagella, and encyst until they produce germ tubes (Figure 13.1A, B). This occurs after approximately two hours at an optimum temperature (12 to 15° C). Germ tubes differentiate into appressoria that invade the host through the leaf cuticle, or less frequently, the stomata (Figure 13.1A, B). The optimal germ tube differentiation temperature is between 21 and 24° C. Intercellular hyphae develop and travel inside the host between cells, using haustoria to form biotrophic feeding relationships in the mesophyll (Figure 13.1A, B). Rapid colonization occurs between 22 and 24° C. Hyphae spread and sporangiophores eventually emerge from stomata (Figure 13.1C, D). Soon thereafter, LB symptoms become apparent, generally between five and ten days after inoculation. Sporulation occurs to produce 2N sporangia, which eventually release zoospores to promote aerial transmission of LB and continue the disease cycle (Judelson 1997). Disease development ceases if temperatures increase above 35° C, although *P. infestans* can survive in living host tissue and the disease can progress when conditions again become favorable (Scott and Gardner 2007).

Sporangia develop many organelles that are absent in hyphae, such as large peripheral vesicles, encystment vesicles, kinetosomes, and flagella. Zoosporogenesis involves cleavage of the multinucleate sporangial cytoplasm by nucleus-enveloping membranes. Subsequently, assembly of flagella, dissolution of the sporangial papilla, and expulsion of uninucleate zoospores take place. Several of the early-induced genes encode participants in vesicle movement that may aid in assembling these structures. Other early-induced genes encode components of the spore-specific vesicles or organelles themselves, including more than 70 flagella-associated, as well as a thrombospondin-like encystment proteins. Under favorable field conditions, sporangia can maintain viability for as long as a week, although levels of mRNAs enabling germination and affecting its dynamics usually decrease during this period (Judelson 1997; Hardham and Blackman 2010; Nowicki et al. 2012).

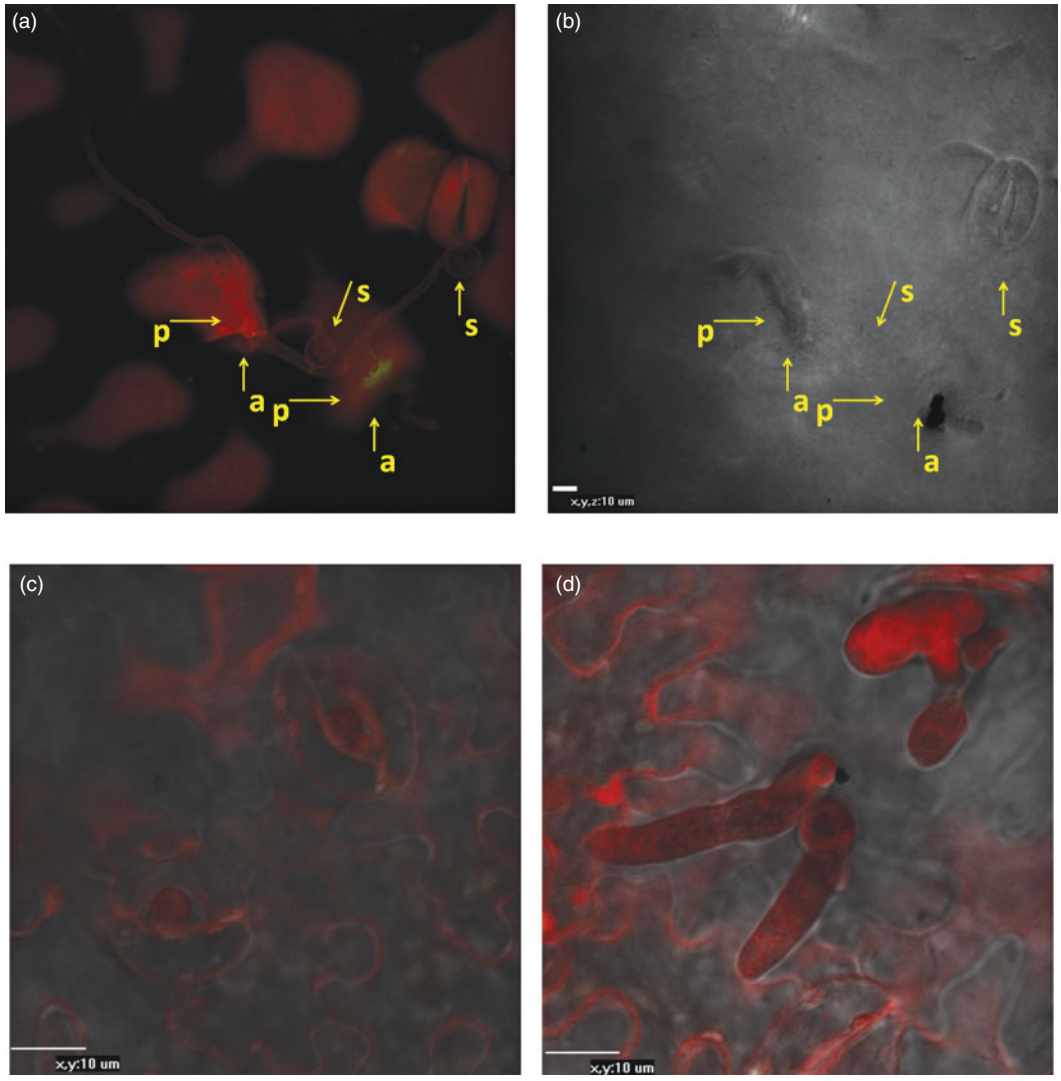


Fig. 13.1. *Phytophthora infestans* infection cycle on susceptible tomato cv. Rumba, as visualized by dual stain and confocal laser microscopy (Nowicki et al. 2012). Red channel: pathogen-specific stain and plant autofluorescence; green channel: callose-specific stain; black and white: DIC. (a) Germinating spore [s] develops infection tube and attempts leaf tissue penetration by formation of appressorium [a]. Host plant's primary line of defense is rapid formation of callose-rich papilla [p] directly underneath the attempted penetration area. (b) Same as A, but visible in DIC. (c) Upon successful infection, the pathogen grows in the leaf mesophyll, as denoted by the numerous hyphae. (d) At the end of the infection cycle, the hyphae emerge through the abaxial leaf-side stomata, followed by sporangial development. Rapid field infestation is achieved through successful colonization; descendant sporangia will drive the epidemics through the season. For a color version of this figure, please refer to the color plate.

The sexual life cycle of *P. infestans* requires the mating of individuals with opposite mating types, known as A1 and A2 (Gallegly and Galindo 1958; Gisi et al. 2011). The mating types are not dimorphic forms of *P. infestans*, but are compatibility types differentiated by mating hormones (Judelson 1997). When mycelia of the two mating types interact, mating hormones induce gametangial formation in the opposing mating types, resulting in sexual reproduction. In

the course of gametangia formation, vegetative, diploid mycelia undergo meiosis to form haploid antheridia and oogonia. During the sexual life cycle, an antheridium fuses with an oogonium to form a diploid oospore. Unlike sporangia, which are airborne, fragile, and need live plants for survival, oospores are large, thick-walled spores that enable pathogen survival for extended periods of time in harsh conditions outside the living host plant. Oospore survival in plant debris or soil outside the living host can serve as a persistent source of inoculum in the field, similar to potato tuber-borne inoculum (Nowicki et al. 2012). Oospores can germinate under environmentally favorable conditions and release diploid progeny of A1 or A2 mating type (Judelson 1997). Sexual recombination occurs in the presence of both mating types, giving rise to new and possibly more aggressive isolates, thus making LB disease management more challenging (Fry and Goodwin 1997; Fry 2008).

There are several reasons LB remains a major threat to Solanaceae crop species. As stated above, *P. infestans* reproduces very rapidly and can destroy an unprotected tomato or potato crop within several days of occurrence. The asexual disease cycle consisting of pathogen penetration, colonization, sporulation, and dispersal can occur in fewer than five days. Each LB lesion can produce as many as 300,000 sporangia per day (Figure 13.1C, D), contributing to a rapid spread of the disease. Further, plants of a given crop are not simultaneously affected, making the early stages of disease easy to miss. By the time the disease is detected it is often too late to save the crop through fungicide application.

Prior to the LB resurgence (see below), only the A1 mating type was observed in potato growing areas outside Mexico, so sexual reproduction did not play a significant role in the disease cycle. Migration of the A2 mating type outside Mexico in the 1980s created the opportunity for pathogen sexual reproduction and DNA recombination, resulting in the creation of new, more aggressive isolates (Fry and Goodwin 1997; Fry 2008). The resultant oospores enable the organism to survive for long periods in plant debris or soil outside the living host plant (Judelson 1997), and may play a key role in the epidemiology of LB and serve as a persistent source of inoculum in the field, similar to potato tuber-borne inoculum.

The final concern associated with LB reemergence is that many new, complex *P. infestans* lineages exhibit resistance to metalaxyl, the predominant systemic fungicide used to control the pathogen. Improper applications of phenylamide fungicides have created a selective pressure on the pathogen, leading to the establishment of phenylamide-resistant isolates of *P. infestans* in many regions throughout the world. The occurrence of metalaxyl resistance nearly coincided with the observation of the A2 mating type outside Mexico, however, no genetic correlation has been documented between mating type and metalaxyl resistance (Gisi and Cohen 1996). Metalaxyl resistance poses a great threat to tomato and potato growers, as these are the main systemic fungicides against LB. The presence of metalaxyl resistance suggests that once LB is observed, it is likely to be too late to use protectant fungicides to save the crop. Despite different means to circumvent this issue (e.g., distribution of these fungicides only in combination with at least one other fungicide with a different mode of action, or metalaxyl replacement by its optical isomer, metalaxyl-m [mefenoxam]), the concern with fungicide resistance further stresses the need for identifying sources of host genetic resistance against the pathogen (discussed below).

P. infestans Pathogenesis

The life cycle of *P. infestans* involves differentiation into as many as 11 different cell types. These cell types are highly specialized for life cycle stages involved in sexual and asexual reproduction, propagule dispersal, spore germination, host penetration, and biotrophic or necrotrophic phases of infection (Nowicki et al. 2012). *P. infestans* adopts a two-step infection mode, typical of hemibiotrophs. During the early

biotrophic growth phase, nutrients are obtained from living plant cells. This is achieved by forming a penetration peg, which pierces the cuticle and enters an epidermal cell to form an infection vesicle (Figure 13.1A, B). Formation of digit-like haustoria (i.e., appressoria-like structures) by the branching hyphae, and the subsequent secretion of enzymes that degrade components of the plant cell wall further facilitate the nutrient access. Haustoria, specialized for nutrient uptake from the host cell, develop within the boundary of the plant cell wall, although they remain outside the host cell plasma membrane (Hardham and Blackman 2010). This is followed by extensive necrosis of host tissue resulting in colonization and sporulation. During necrotrophic growth, nutrients required for pathogen growth and reproduction are obtained from dead and dying cells in the necrotic lesions that develop as the pathogen colonizes the plant. As the infected tissue necrotizes, the mycelium develops sporangiophores that emerge through the stomata to produce numerous asexual sporangia (Hardham and Blackman 2010; Nowicki et al. 2012) (Figure 13.1C, D).

P. infestans preferentially sporulates *in planta* during dark periods. Studies have demonstrated that continuous light inhibits sporulation (Nowicki et al. 2012). As there is no evidence for a light-regulated clock in the pathogen (Raffaele et al. 2010b), this suggests that a plant-derived signal generated by the day-night cycle influences sporulation. Similar phenomena are common among the oomycetes (Judelson et al. 2009; Hardham and Blackman 2010). At the plant surface, *P. infestans* zoospores maneuver such that their ventral surface faces the plant before the flagella are detached and proteins from three different cortical vesicles are secreted onto the zoospore surface (Hardham and Blackman 2010). By doing so, the spores form walled cysts, which germinate from the center of the ventral surface, allowing the hyphae to grow along the anticlinal walls between epidermal cells to form appressoria for breaching plant cells (Figure 13.1A, B). Under favorable conditions, the multinucleate sporangia can also initiate infections. Mature sporangia can preemptively synthesize the proteins involved in zoosporogenesis and encystment (Nowicki et al. 2012).

In general, during infection, phytopathogens secrete peptides and proteins, broadly known as effectors (Raffaele et al. 2010b; Vleeshouwers et al. 2011). Effectors can be of two types: apoplastic effectors, which target the extracellular space, and cytoplasmic effectors, which target subcellular compartments. Effectors include enzymes involved in degrading the plant cell wall and suppressing extracellular plant defenses (Vleeshouwers et al. 2011; Hardham and Blackman 2010; Nowicki et al. 2012). In-depth discussion of these proteins in *P. infestans* pathogenesis is presented in the potato LB chapter (Chapter 12 in this volume), as major advances in this field have been achieved in research on this host plant.

Disease Symptoms and Progression

P. infestans can quickly devastate tomato and potato crops at any time during plant ontogeny. The entire plant may collapse in five to ten days. LB can infect all aboveground parts of the plant, causing leaf and stem necrosis, fruit rot, and eventual plant death (http://www.nysipm.cornell.edu/publications/blight/). The pathogen can also infect tomato seed and potato tubers (Rubin et al. 2001; Nowicki et al. 2012). Initial infection symptoms, including small lesions on leaf tips and plant stems, are visible only after three to four days, and in some cases reach only 1 to 2 mm in diameter. The purple, dark brown, or black water-soaked lesions often have a pale yellowish-green border that blends into the healthy tissue. As the pathogen penetrates the plant tissue, lesions enlarge in size (http://www.longislandhort.cornell.edu/vegpath/photos/lateblight_tomato.htm). Fluffy, white sporangia may grow on the lower (abaxial) leaflet surface in moist weather. As the disease progresses, plant leaflets shrivel and die and the disease spreads to the rest of the foliage, leading to extensive defoliation. Dark brown LB lesions first appear at the

top of the stem or at a node and may progress down the stem. Firm, brown, and greasy tomato fruit lesions are often located at the stem end and sides of green fruit, rendering them unmarketable. Infected tomato fruit may be invaded by secondary pathogens, causing soft-rot disease.

Host Plant Defense

Plants are finely tuned to detect the presence of pathogens on their surface. They perceive both chemical and physical signals of pathogens and react rapidly to the attempted infection (Hardham and Blackman 2010; Nowicki et al. 2012). Possible plant responses to oomycete invasion vary from invaded cells with no visible response, to more localized reactions preventing formation of pathogen intra-cellular structures, to hypersensitive response (HR) and programmed cell death, the latter occurring only after pathogen growth had proceeded to the formation of a clearly identifiable haustorium (Hardham and Blackman 2010; Hardham et al. 2007).

The primary line of induced defense in tomato plants against *P. infestans* is resistance against penetration at the leaf surface. This is a highly effective defense strategy rapidly mobilized by host plants to attempted penetrations (Hardham et al. 2007). Recent studies have demonstrated the direct participation of plasma membrane-bound receptor proteins in the recognition of apoplastic pathogen elicitors and activation of host defenses (Haas et al. 2009). Additionally, the physical pressure exerted by the invading penetration peg may act as a signal to plant surface detection systems. Triggering of this system normally leads to reversible formation of microfilaments beneath the pressure point within minutes of the acting stimulus. Both stimuli may set off the downstream signaling pathways, including salicylic acid-regulated induction of cell death, ethylene-dependent production of phytoalexin, and reactive oxygen species (ROS)-regulated gene expression and induction of HR (Haas et al. 2009; Shibata et al. 2010; Nowicki et al. 2012). In response to the perceived pathogen attack, plant cells secrete a broad spectrum of compounds to combat the disease, including constitutively expressed and inducible defense-related proteins. These include defense-related proteins, which function in the degradation of microbial cell walls and inhibiting pathogen-released elicitors (Foolad et al. 2008; Hardham and Blackman 2010). Early localized epidermal cell reactions include major rearrangements within the cytoplasm leading to its aggregation beneath the pathogen site, synthesis of ROS, and accumulation of callose beneath the appressoria within 14 hours after inoculation or natural pathogen attack (Hardham et al. 2007; Hardham and Blackman 2010) (Figure 13.1A, B). The rapidly occurring site-directed cytoplasmic streaming is achieved by reorganization of cortical and vascular actin arrays, such that actin microfilaments and cables form a radial array focused on the infection site. Subsequently, because of the actin-dependent focus of cellular secretory apparati (endoplasmic reticulum, Golgi bodies, and peroxisomes) beneath the infection site, a wide variety of molecules are delivered to the developing cell wall apposition (Hardham et al. 2007; Hardham and Blackman 2010). Callose, a major component of the cell wall apposition (Figure 13.1A, B), alongside other cell wall apposition components, such as phytoalexins, phenolics, silicon, H_2O_2, and enzymes such as peroxidases and enzyme inhibitors, is likely to inhibit pathogen hyphal penetration. Subsequent defense responses include changes in gene expression and production of phytoalexins and pathogenesis-related proteins (Hardham et al. 2007; Shibata et al. 2010).

Only after primary host-plant defenses fail and hyphae successfully penetrate with the formation of haustorium does the plant invoke a second system of resistance against *P. infestans*, which involves recognition of specific pathogen molecules and induction of programmed cell death. HR constitutes a highly effective means of restricting pathogen growth and its containment before transition to the necrotrophic phase

occurs (Vleeshouwers et al. 2000; Hardham et al. 2007). Elicitins and effectors are the two main groups of molecules that induce HR. Upon their being recognized by host plants, elicitins induce HR. For instance, *P. infestans* INF1 elicitin initiates cell death upon interaction with the lectin-like receptor kinase NbLRK1 in *N. benthamiana* (Hardham et al. 2007; Kanzaki et al. 2008). In total, genes coding for seven avirulence effector proteins of *P. infestans* have been cloned at present (Vleeshouwers et al. 2011; van Damme et al. 2012), and all of them are recognized by cytoplasmic nucleotide-binding site/leucin-rich repeat (NBS-LRR) receptors (Jia et al. 2010; Vleeshouwers et al. 2011) encoded by the plant resistance (*R*)-genes. Race-specific and broad-spectrum *R*-genes in host plants induce a similar set of defense reactions as a response to recognition of their corresponding *P. infestans* avirulence factors, the effector proteins (Shibata et al. 2010; Nowicki et al. 2012). In tomato, a wide range of proteins secreted by *P. infestans* and the host plant during a compatible interaction was identified in an *in planta* secretome study using the yeast secretion trap technique (Lee et al. 2006). Almost half the secreted proteins in the tomato plants had known associations with defense, including pathogenesis-related (PR) proteins, structural proteins, glycosyl hydrolases (which target glucans, typical constituents of microbial cell walls), and a putative peroxidase. Other secreted proteins had less defined roles in defense, including those induced by wounding or elicitors. Additionally, about one third of the induced genes had no obvious functional domains or homology to known genes (Lee et al. 2006).

To date, no mechanisms for LB resistance have been proposed in either tomato or potato. A preliminary study was conducted to elucidate the mechanism of vertical LB resistance conferred by the *S. bulbocastanum R*-gene, *RB*, by investigating the effects of *RarI* and *SgtI* on LB resistance (Bhaskar et al. 2008). Unlike the resistance conferred by most major genes, *RB* slows disease progress but does not eliminate disease symptoms (see Chapter 12 of this volume). *RarI* and *SgtI* are known to regulate *R*-genes' expression. Although proposed to be involved in forming or stabilizing R-protein associated recognition complexes, silencing *RarI* using RNAi had no effect on *RB* resistance, indicating that it is not required in the LB-resistance response. Conversely, silencing *Sgt1*, known to be involved in NBS-LRR and Pto-kinase mediated resistance, resulted in disease susceptibility, indicating that SGT1 plays a role in LB resistance (Shibata et al. 2011). Another study on phenotypic characterization of potato LB resistance conferred by *RB* suggests that expression of *HR* and *PR* genes may play a critical role in the resistance response, with callose deposition being negatively correlated with resistance levels in tested plants (Chen and Halterman 2011).

Collectively, multiple defense mechanisms seem to be involved in LB resistance and alteration of metabolic pathways may be one of the most important disease defense responses (Bhaskar et al. 2008; Foolad et al. 2008; Chen and Halterman 2011). Nevertheless, innate tomato-derived LB-resistance mechanisms alone may not suffice to defend the plants against the aggressive pathogen, hence the need to integrate various protection methods into crop production.

Late Blight Disease Control

Tomato and potato LB are of significant fiscal importance to growers and consumers worldwide, costing approximately US$5 billion annually, including the cost of disease control and crop losses (Foolad et al. 2008). In the United States, in 2009 total yield losses for fresh-market and processing tomato industries reached US$46 million and US$66 million, respectively (USDA 2011). Of these losses, up to half could be attributed to crop losses resulting from LB (Haverkort et al. 2008; Nowicki et al. 2012). In tomato, fruit infection may range from 41 to 100% in unprotected fields and from 12 to 65% in plots protected with systemic fungicides (Nowicki et al. 2012). Polish tomato production has

suffered significantly from rampant outbreaks of LB over the past several years, often causing yield losses of up to 100% (Nowicki et al. 2012). Social impact of LB is illustrated by the significant media and internet coverage (Vleeshouwers et al. 2011). Losses resulting from tomato LB are accompanied by higher fungicide applications worldwide, exemplifying the need for crop protection-related expenditures and environmental safety concerns. Estimated cost of fungicides and crop losses resulting from LB in the U.S. exceeds $210 million annually (Foolad et al. 2008). Predictions that fungicide applications needed to control LB in the future would increase by 20 to 25%, compared to that required from 1977 to 2008 (Nowicki et al. 2012), further add to concerns over the fiscal and environmental costs related to LB control.

Prior to the reemergence of LB in the late 1980s, cultural practices in combination with fungicide applications were highly effective measures for controlling LB in tomato. These approaches, however, are not expected to provide sustainable control of the disease in the future, as discussed below. It seems the greatest contribution to tomato LB control in the future will have to be through the development of cultivars with improved genetic resistance via modern plant breeding approaches.

Cultural Practices

Cultural practices are important components of growers' strategy in disease management, and they can impact disease development and control. The aims of cultural control of LB are to minimize inoculum buildup, prevent introduction of inoculum from nearby potato cull piles or from tomato transplants, minimize infection rate, and generate conditions unfavorable for disease development and spread. Specific cultural practices usually employed to control LB include crop rotation and fallow, elimination of volunteer tomato and potato plants, planting noninfected seedlings and tubers, and elimination of LB sources such as potato cull piles. The latter is of particular importance because cull piles can serve as a living host on which *P. infestans* mycelia can survive over the winter and produce tremendous amounts of airborne spores at the beginning of the new field season. If this occurs, the next year's crop is at risk of LB devastation (Foolad et al. 2008). Recent reemergence of *P. infestans* with its enhanced ability to develop more virulent isolates through sexual recombination makes LB control via cultural practices alone very challenging, and the pathogen could be particularly destructive in areas where both tomatoes and potatoes are grown year-round, as for example in the highland tropics of Africa, South America, Asia, and Europe (Nowicki et al. 2012).

Fungicide Application

Chemical control measures can be effective in managing LB, especially when guided by disease forecast systems, and have been increasingly utilized in recent years. Two main groups of fungicides have been used routinely, including protectants (e.g., chlorothalonil, dithiocarbamates, and triphenyl tin hydroxide), which are usually applied before or upon disease development, and systemic fungicides (a.k.a. therapeutic fungicides; e.g., phenylamides such as metalaxyl/mefenoxam, aliphatic nitrogen fungicides such as cymoxanil, and morpholine fungicides such as dimethomorph), which inhibit or reduce disease progress once symptoms are apparent. Current chemical practices to control LB include a mixture of fungicides designed to slow the disease progress. Metalaxyl fungicides, a class of systemic fungicides, have been widely employed to control LB; they inhibit ribosomal RNA (rRNA) polymerases in fungi by reducing incorporation of uridine (Gisi et al. 2011). These treatments, however, can be ineffective, particularly when the environmental conditions are highly conducive to disease development. Moreover, improper applications of phenylamides have created a selective pressure on the pathogen, leading to the spread of fungicide resistance controlled

by a single, incompletely dominant gene present in natural *P. infestans* populations. Establishment of metalaxyl-resistant isolates of *P. infestans* was observed in many countries as early as 1980 (Gisi et al. 2011). In addition to becoming increasingly ineffective as a result of the development of pathogen resistance, fungicides are expensive, harmful to the environment and humans, and must be applied at the proper time. With the use of at least partially resistant cultivars, the number of fungicide applications and/or the rate of application can be significantly reduced, particularly when combined with LB forecasting (Foolad et al. 2008). It should be noted that copper fungicides have also been used in organic fresh-market tomato production, but they have been shown to only suppress the LB symptoms, while not stopping the disease. In a 2012 field investigation of the effects of different fungicide programs on controlling tomato LB, the organic fungicide program consisting of copper hydroxide tank mixed with *Bacillus subtilis* only helped reduce LB infection but did not control the disease (Gugino and Foolad, unpubl. data). Overall, the issues surrounding the use of fungicides, as well as economic and environmental safety considerations, necessitate careful adoption of effective and sustainable disease control measures, including the development and integration of commercially acceptable cultivars with genetic resistance against LB. The more resistant the cultivar, the greater its potential for reduction of fungicide application (Foolad et al. 2008).

Genetic Resistance against LB in Tomato and the Importance of Breeding for Resistance

Since the Irish famine of the 1840s, there has been a great deal of interest in developing LB-resistant potato and tomato cultivars. In the course of this process, it has been observed that resistance against *P. infestans* could be classified into race-specific resistance (a.k.a. pathotype-specific, vertical, or "gene-for gene" interaction) and race-non-specific resistance (a.k.a. field, horizontal, or partial resistance), with races defined by the disease interaction with different host plant genotypes. In tomato, similar to potato, vertical as well as horizontal resistances against LB have been reported. Developing plants that display disease resistance conferred by major genes (i.e., vertical resistance) have been effective (at least initially) in preventing infection and protecting the crop from LB (see below). The host resistance gene product, also known as the *R*-gene product, interacts with the pathogen's pathogenicity gene product, also known as the *Avr*-gene product. Single gene resistance typically confers complete resistance against one or a limited number of pathogen races. Due to rapid evolution of pathogen effectors and sexual reproduction of *P. infestans* leading to more aggressive lineages, vertical resistance against the pathogen could ultimately fail. For example, durability of major LB-resistance genes in potato has proven variable (Vleeshouwers et al. 2011), and isolates of *P. infestans* have been identified that basically overcome all 11 *R*-genes identified in potato wild species *S. demissum* (Chen and Halterman 2011). Similar situation may be stipulated for major LB-resistance genes in tomato (Nowicki et al. 2012), though this has not been documented (discussed below).

In contrast to vertical resistance, race-non-specific resistance is often controlled by several genes or quantitative trait loci (QTLs), and potentially could be more durable. Horizontal resistance usually confers partial resistance against multiple isolates/races of the pathogen. This type of resistance often slows, but does not stop progress of the disease. The polygenic nature of horizontal resistance makes it more difficult to breed for, compared with vertical resistance, however, its greater durability may render the breeding efforts worth undertaking. In the case of tomato LB, it has been argued that horizontal resistance is likely of limited value because of the pathogen's short reproductive cycle and heavy spore production, as well as the ability of the disease to spread quickly

by airborne spores over long distances (Scott and Gardner 2007). Conversely, race-specific major resistance genes have been useful in breeding tomatoes for resistance against LB, and in fact against many other tomato diseases (Foolad 2007). Further, many major resistance genes have shown durability and been very useful for breeding purposes in other crop species. The use of major disease resistance genes, however, has been shown to be most effective when multiple resistance genes are pyramided; this approach often increases resistance strength and durability (Kim et al. 2012; Luo et al. 2012). Thus, it is prudent to identify additional major resistance genes against tomato LB, a task that is underway in many tomato-breeding programs around the world (Merk and Foolad 2011, Merk et al. 2012; Nowicki et al. 2012).

Identification of Resistance Resources and Breeding for LB resistance in Tomato

Overview of Early Studies of LB Resistance

Following an LB outbreak in the U.S. in 1946, which affected potatoes and tomatoes, a substantial amount of research was initiated to locate sources of genetic resistance in tomato (Bonde and Murphy 1952; Gallegly and Marvel 1955; Gallegly and Galindo 1958; Gallegly 1960). Such research led to the discovery of resistant accessions within wild tomato species, in particular *S. pimpinellifolium* (Gallegly and Marvel 1955). The first reported tomato LB-resistance gene, *Ph-1*, a completely dominant gene conferring resistance against *P. infestans* tomato race-0 (T_0), was originally located in *S. pimpinellifolium* accessions known as West Virginia 19 and 731 (Bonde and Murphy 1952; Gallegly and Marvel 1955; Pierce 1971). An LB-resistant cultivar containing *Ph-1*, Rockingham, was released in 1962 (Rich et al. 1962). This cultivar was subsequently used to map the gene conferring LB resistance to the distal end of chromosome 7 using morphological markers (Pierce 1971). Subsequently this resistance trait was incorporated into the old processing tomato cv. 'Nova' and the old fresh market tomato cv. 'New Yorker'. Since T_0 is no longer the predominant race of *P. infestans*, and since *Ph-1* has been long overcome by new aggressive pathogen lineages, this LB-resistance source is no longer considered useful for tomato breeding (Foolad et al. 2008; Nowicki et al. 2012). Currently, cultivars containing *Ph-1* exhibit complete susceptibility to LB.

A second tomato LB-resistance gene named *Ph-2* was identified in an *S. pimpinellifolium* accession known as West Virginia 700 (Gallegly and Marvel 1955) and subsequently mapped to the long arm of tomato chromosome 10 between markers CP105 and TG233 (Moreau et al. 1998) (Figure 13.2A). There has not been any further effort to fine map or clone *Ph-2* (N. Grimsley, CNRS-INRA, personal communication). Incompletely dominant LB resistance conferred by *Ph*-2 provides only partial resistance against several pathogen isolates and confers only a reduction in the rate of disease development, rather than blocking the disease. Furthermore, *Ph-2* often fails in the presence of more aggressive *P. infestans* isolates. Characterization of this resistance has been hampered because its expression is partially dependent upon environmental conditions, plant physiological age, the organ assessed, and the pathogen isolate used (Moreau et al. 1998). Despite these shortcomings, *Ph-2* has been successfully incorporated into a number of named fresh-market and processing tomato varieties, including Legend, Centennial, Macline, Pieraline, Herline, Fline, Flora Dade, Heinz 1706, Campbell 28, and Europeel (Gallegly 1960; Nowicki et al. 2012). Recently a few PCR based markers associated with *Ph-2* have been identified (M. Mutschler, personal communication), which are being used for development of breeding lines and hybrid cultivars of tomato containing *Ph-2*. Nevertheless, virulence variability observed in *P. infestans* has made this source of tomato resistance less

effective for protection against LB when used alone (see below).

Resurgence of LB and Identification of Additional Resistance Genes and QTLs

Until the late 1970s, LB was relatively well controlled through the use of cultural practices, heavy application of fungicides, and deployment of somewhat resistant cultivars of potato. LB reemerged in the 1980s as an important disease of both potato and tomato. This has been in part because of several independent global migrations of the pathogen from its Andean origin (Fry and Goodwin 1997) and potentially the occurrence of sexual reproduction in populations of *P. infestans*. The latter has resulted in generation of new and more aggressive isolates of the pathogen, many of which exhibit resistance to known systemic fungicides (Gisi and Cohen 1996; Gisi et al. 2011). After several years of observation, it has become clear that these new isolates of the pathogen constitute a greater threat to potato and tomato production than did the previous dominant lineage, US-1. The new *P. infestans* lineages differ in pathogenicity factors (specific virulence, aggressiveness, and fungicide specificity) and ecological factors (responses to physical factors such as temperature; Fry and Goodwin 1997). Moreover, host preference change has been observed. For example, prior to the 1980s many of the potato isolates were not pathogenic to tomato, whereas new isolates with greater pathogenicity to tomato have appeared since (Fry 2008). During the 1980s and 1990s, perhaps the most significant consequence with respect to tomato LB was the breakdown of resistance conferred by *Ph-1* and *Ph-2* in tomato. Such a breach of resistance as well as the appearance of new and more aggressive isolates of *P. infestans* resulted in worldwide increased occurrence of LB in tomato (Fry and Goodwin 1997), reaching as far as Taiwan, Nepal, Indonesia, and the Philippines. This prompted further screening of tomato wild species, in particular accessions within *S. pimpinellifolium*, to find new sources of LB resistance. Subsequent research led to the discovery of a strong LB-resistance gene named *Ph-3* in *S. pimpinellifolium* accession L3708 (a.k.a. LA1269 and PI 365957) (AVRDC 1993).

Ph-3 confers resistance against a wide range of *P. infestans* isolates that overcome *Ph-1* and *Ph-2*. A bulked segregant analysis (BSA) using AFLP (amplified fragment length polymorphism) markers resulted in the identification of DNA markers associated with *Ph-3* and enabled mapping of *Ph-3* to the long arm of chromosome 9, close to RFLP (restriction fragment length polymorphism) marker TG591A (Chunwongse et al. 2002) (Figure 13.2B). Subsequently a number of DNA markers, including RFLP- and PCR-based markers, closely linked to *Ph-3* have been reported (Zhu et al. 2006; Qiu et al. 2009; Park et al. 2010; M. Mutschler, personal communication) (Figure 13.2B). Recently several public tomato breeding programs in the U.S., including programs at North Carolina State University (R.G. Gardner and D.R. Panthee, personal communication), Cornell University (Kim and Mutschler 2005; M. Mutschler, personal communication), and the Pennsylvania State University (M.R. Foolad unpublished results), have developed fresh-market and/ or processing tomato lines possessing *Ph-3*. It has been determined, however, that LB resistance in L3708 is controlled by more than just the *Ph-3* locus on chromosome 9 and that LB-resistant breeding lines developed at the Asian Vegetable Research and Development Center, in Taiwan (AVDRC) were probably lacking at least one of those additional hypostatic genes and were not as resistant against LB as was L3708 (Kim and Mutschler 2005; Lee et al. 2006). The authors further concluded that inbred lines or hybrids that contain the *Ph-3* gene alone in either homozygous or heterozygous condition would not be highly desirable commercially, as they would not exhibit strong resistance against aggressive isolates such as US-7 and US-17. The presence of the yet undetermined additional hypostatic gene(s) in homozygous or heterozygous

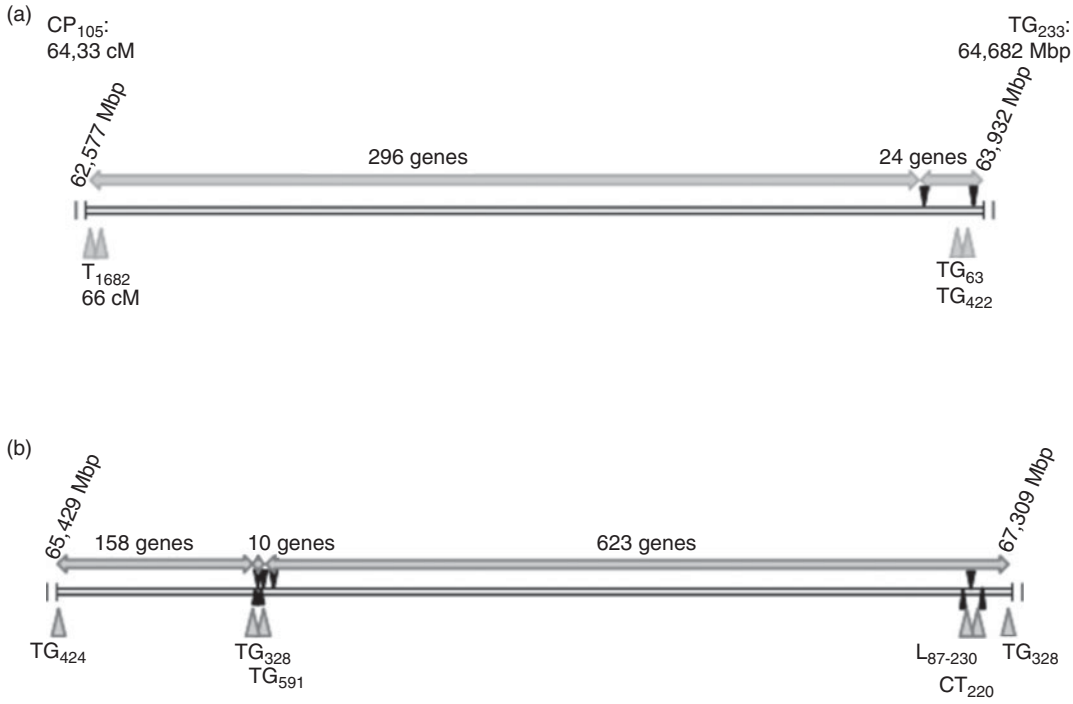


Fig. 13.2. Partial physical maps of tomato chromosomes 10 and 9, to which tomato late blight resistance genes have been mapped. Red triangles indicate the approximate positions of respective markers and black arrowheads indicate the approximate positions of *NBS-LRR* genes present in the two regions. (a) A fragment of chromosome 10 to which *Ph-2* has been mapped between markers CP105 at 64.33 cM and TG233 at 64.82 cM (Moreau et al. 1998; http://solgenomics.net/); the fragment is narrowed down to 320 genes between markers T1682 (Marczewski, Nowakowska, Nowicki, Kozik unpublished results) and TG63 and TG422 (Foolad and Panthee 2012). (b) A fragment of chromosome 9 to which *Ph-3* has been mapped (Chunwongse et al. 2002); this fragment covers 791 genes and contains markers reported in the literature to be associated with *Ph-3* (see Park et al. 2010 and Foolad and Panthee 2012).

condition is necessary to provide full resistance (Kim and Mutschler 2005). Nevertheless, despite the superiority of the *Ph-3* complex, as compared with *Ph-1* or *Ph-2*, *P. infestans* isolates have emerged that overcome this resistance (Chunwongse et al. 2002; AVRDC 2005; R.G. Gardner, personal communication).

It has been observed, however, that Ph-3 is most effective when combined with *Ph-2* (R.G. Gardner, personal communication; M.R. Foolad unpublished results). Most recently, these two resistance genes have been pyramided in a few tomato breeding lines (e.g., NC1 CELBR, NC2 CELBR) and hybrid cultivars (e.g., Mountain Magic and Mountain Merit), which exhibit much stronger LB resistance than breeding lines or cultivars that have only *Ph-2* or *Ph-3* (Gardner and Panthee 2010a; Panthee and Gardner 2010; MR Foolad, pers. observations). Scott and Gardner (2007) reported, however, that under ideal environmental conditions for LB, including heavy inoculum pressure and presence of other lines with single gene resistance in the same plantings, the lines with pyramided *Ph-2* and *Ph-3* genes in either homozygous or heterozygous state were overcome by LB in two field locations in North Carolina, USA, in the summer of 2004. In contrast, in the absence of lines carrying either *Ph-2* or *Ph-3* resistance genes alone, the resistance in lines or cultivars with both *Ph-2* and *Ph-3* held up throughout the season (Scott and Gardner 2007). In comparison, in a 2012

multi-location trial of numerous tomato genotypes containing *Ph-2* and/or *Ph-3* in Pennsylvania, USA, it was determined that lines and cultivars containing both *Ph-2* and *Ph-3* exhibited very strong resistance under high LB pressure throughout the season (M.R. Foolad and B.K. Gugino unpublished results).

In addition to the abovementioned major LB-resistance genes identified in tomato wild species *S. pimpinellifolium*, several quantitative trait loci (QTLs), potentially conferring race-non-specific resistance, have been reported in different accessions of the green-fruited tomato wild species *S. habrochaites*. The first accession of *S. habrochaites* discovered with LB resistance was LA1033 (Lough 2003), and the LB-resistance gene/QTL it carried was designated by AVRDC as *Ph-4* (AVRDC 1993; 2005). In *S. habrochaites* accession LA2099, QTLs conferring LB resistance were identified on all 12 tomato chromosomes (Brouwer et al. 2004). In subsequent research, three near-isogenic lines (NILs) were developed, each containing one of three major LB-resistance QTLs (Brouwer and St Clair 2004). Unfortunately, severe linkage drag has thus far prevented the NILs from being useful for breeding purposes. Further inspection of these NILs has determined that they also contain genes/QTLs for other characteristics, including plant type and maturity, canopy density, and fruit size and yield in the same introgressed regions. In these accessions, QTLs for LB resistance were associated with late maturity and large plant size (Brouwer and St Clair 2004). In a more recent study, five LB-resistance QTLs were identified in *S. habrochaites* accession LA1777 and one new QTL for LB resistance in LA2099 (Li 2010). Nevertheless, the actual value of these QTLs for breeding purposes is yet unknown. Until these QTLs are refined and NILs are developed with individual QTLs in the background of the cultivated tomato, no judgment can be made as to their breeding value. Furthermore, it is important to determine whether the combination of major LB-resistance genes, such as *Ph-2* and *Ph-3*, with LB-resistance QTLs would increase the level or durability of resistance against LB. Although it is difficult to predict the outcome, it would be a worthwhile effort to develop such gene combinations. Regardless, we are not aware of any tomato program that is conducting such a research at this time.

Recent Screening and Identification of New LB-Resistance Genes

The recent occurrence of newer, more aggressive isolates of *P. infestans*, which have been highly pathogenic to tomato, necessitated a search for further sources of LB resistance in tomato. During the past several years, a comprehensive effort towards identification, mapping, and pyramiding LB-resistance genes has been undertaken at the Pennsylvania State University. This effort was initiated with screening of a large collection of *S. pimpinellifolium* accessions for LB resistance under different conditions, including field, high tunnel, greenhouse, and growth chamber conditions. The screening trials resulted in the identification of several new accessions with strong resistance against tomato LB (Foolad et al. 2008; M.R. Foolad unpublished results). Among the newly identified accessions, one accession (PI 270443) that exhibited strong resistance against at least 7 isolates of *P. infestans* was chosen for further characterization, including inheritance studies and identification and mapping of resistance gene(s). A parent-offspring correlation analysis, employing F_2 and F_3 populations of a cross between PI 270443 and a susceptible tomato breeding line (NC EBR-2), determined that the LB resistance conferred by PI 270443 was highly heritable ($h^2 = 0.86$) and estimated to be controlled by two genes (Merk and Foolad 2011). Subsequently, a selective genotyping approach, using various molecular markers (RFLP, SSR, EST, CAPS), led to the identification of two genomic regions on tomato chromosomes 1 (tentatively named *Ph-5-1*) and 10 (tentatively named *Ph-5-2*) associated with LB resistance in PI 270443 (Merk et al.

2012). The genomic location on chromosome 10 (*Ph-5-2*) is co-localized with *Ph-2*, however, it is unknown whether it is the same as *Ph-2*, an allele of the same gene, or a separate but tightly linked resistance gene. Currently, efforts are underway to further delineate the two regions on chromosomes 1 and 10 associated with LB resistance (M.R. Foolad unpublished results). Furthermore, LB resistance from PI 270443 is being introduced into Penn State fresh-market and processing tomato lines by a combination of MAS and traditional breeding protocols (M.R. Foolad unpublished results). Once the individual value of these new genes is determined, it is imperative to combine them with *Ph-2* and/or *Ph-3* to determine whether such gene combinations would increase the strength or durability of tomato resistance against LB. In particular it is important to determine whether combinations of such genes increase the resistance against additional *P. infestans* isolates.

Other Challenges in Breeding for LB Resistance in Tomato

Despite encouraging progress toward overcoming the low genetic diversity of the cultivated tomato, which has been achieved through frequent gene introgressions from the related wild species of tomato, there are other concerns related to LB-resistance breeding. Among them, for example, is the lack of clear standards within the cultivated tomato for *P. infestans* race assessment. Researchers have tried to circumvent this difficulty by determining various characteristics of the pathogen isolates, including mating type, fungicide sensitivity, isozyme genotype, and mitochondrial and nuclear DNA fingerprints (reviewed in Nowicki et al. 2012; also see Chapter 12). Furthermore, some researchers have tried to characterize/classify *P. infestans* isolates based on their pathogenicity on established resistant lines carrying known LB-resistance genes, which to some extent is analogous to the Black's standards in potato. For example, tomato researchers at the AVRDC started analyzing their *P. infestans* isolate collections in this manner in the early 1990s, when *S. pimpinellifolium* accession L3708 was believed to carry a single resistance gene, *Ph-3*, and *S. habrochaites* accession LA1033 was designated as the source of *Ph-4* resistance gene (AVRDC 1993; 2005). Follow-up investigations revealed complex resistance involving multi-genes in L3708 (Kim and Mutschler 2005) or multi-QTLs in LA1033 (Lough 2003). Classifications such as these constituted a mere first step toward development of a more comprehensive and useful classification system of the races of tomato-virulent *P. infestans* strains. With the availability of *P. infestans* genomic sequence data and identification of additional sources of LB resistance in tomato, more information regarding the various races of the pathogen may soon become available. It should be noted, however, that if pyramided major LB-resistance genes hold effectively, knowledge of races may not be critical from a breeding perspective.

Implications of Potato LB Resistance Research in Tomato

As discussed above, LB resistance, both vertical and horizontal, has been reported in several related wild species of potato. For example, at least 11 race-specific LB-resistance genes (*R1* to *R11*) have been identified in potato wild species *S. demissum*. Several of these have been mapped or cloned and incorporated into various potato cultivars, as discussed elsewhere (Foolad et al. 2008; Nowicki et al. 2012). Currently, most new cultivars of potato carry one or more LB-resistance genes and demonstrate a good level of field resistance. All of the hitherto cloned *R*-genes encode the NBS-LRR class of plant resistance proteins (Jia et al. 2010; Vleeshouwers et al. 2011) and confer race-specific HR responses. It is noteworthy, however, that such race-specific resistance genes may lose their effectiveness as new and more aggressive *P. infestans* strains appear that could overcome the resistance. For example, isolates of

P. infestans have been identified that overcome all 11 *S. demissum*-derived R-genes in potato (Chen and Halterman 2011). Nonetheless, pyramiding various race-specific resistance genes often results in stronger and more durable resistance, as has been observed in many plant species, including potato and tomato (Tan et al. 2010; Kim et al. 2012; Luo et al. 2012).

Pyramiding of LB-resistance genes using conventional breeding approaches may not always be possible or it may be laborious and time-consuming. An alternative approach to introducing single or multiple LB-resistance genes is genetic transformation. This approach, however, requires prior knowledge of the gene(s), including identification, cloning, and characterization. In this respect, a recent strong body of evidence that tomato and potato share the conserved pathway for LB signaling is very beneficial for LB-resistant tomato breeding (Jia et al. 2009; Faino et al. 2010; Jia et al. 2010). The conserved LB-signaling pathway phenomenon has been demonstrated by several research groups (see below), which heterologously over-expressed the cloned potato LB *R*-genes in tomato plants and showed increased LB resistance in the transgenic plants.

An attempt with the transgenic approach using potato *R1* gene heterologously expressed in tomato indicated the activation of the transgene in the isolate-specific, *R1*-incompatible reaction during *P. infestans* attack (Faino et al. 2010). Increased expression of the transgene, but also of the native tomato *PR-1* gene, has been recorded as early as three hours post infection (hpi); additionally, expression of *PR-1* increased by one-fold between three and six hpi in the transgenic plants (incompatible reaction), while in the native tomato plants remained at the three hpi level (Faino et al. 2010). In another study, *Agrobacterium*-mediated transformation and heterologous expression in tomato of the coding sequences of the *R3a*, *R1* (*S. demissum*), or *RB* (*S. bulbocastanum*) genes was employed to obtain transgenic tomato plants (Jia et al. 2009). When challenged with *P. infestans* isolates, the transgenic plants displayed a strong resistance response. The results demonstrated that *R3a* and *R1* conferred resistance against some tomato-virulent isolates, while *RB* granted resistance against all five isolates examined (Jia et al. 2009). Follow-up studies proved that the transgenic tomatoes carrying the potato *R3a* developed HR specific to *P. infestans* strains with the corresponding avirulence gene, *Avr3a* (Jia et al. 2010). This seems to indicate that the signaling pathway from the *R3a-Avr3a* recognition to HR is conserved between potato and tomato. The transgenic tomatoes carrying both *R3a* and *Avr3a* genes, with the latter under the control of a glucocorticoid-inducible promoter, exhibited a localized HR under dexamethasone-induced *R3a-Avr3a* interaction (Jia et al. 2010).

A similar transgenic approach employing novel LB *R*-genes from potato wild species *S. okadae* (*Rpi-oka1*) and *S. mochiquense* (*Rpi-mcq1.1*, *Rpi-cmq1.2*) was employed to enhance LB resistance in potato and tomato cultivars (Jones et al. 2010). In this study, eight of the nine recovered transgenic potato plants of susceptible cultivar Desiree exhibited resistance against the tested *P. infestans* isolates. Furthermore, when cloned under their native promoter and terminator sequences and stably transformed into potato or tomato cultivars, expression of all three genes induced resistance against a wide spectrum of *P. infestans* isolates tested. Interestingly, the spectrum of LB resistance exhibited for each construct was in agreement with that exhibited by the original wild potato accessions harboring these genes. In particular, *Rpi-oka1*–derived LB resistance was retained in the transgenic plants, and the resistance phenotype was not a result of constitutive activation of defense pathways by the transgene (Jones et al. 2010). Similarly, LB susceptible potato and tomato cultivars exhibited resistance when transformed with resistance gene *Rpi-vnt1.1* from *S. venturii* (Foster et al. 2009). In this study, the transgenic potato and tomato plants exhibited resistance against 10 of

the 11 isolates tested; only isolate EC1 from Ecuador was able to overcome RPI-VNT1.1 and cause disease on the inoculated plants (Foster et al. 2009).

Because potato and tomato are closely related and share high genome similarity, the LB-resistance genes and QTLs identified in potato may have relevance in tomato or correspond to tomato LB-resistance genes and QTLs (Foolad 2007; Vleeshouwers et al. 2011). By analyzing the tomato genome regions to which LB R-genes have been mapped and taking into account that all such potato genes thus far cloned possess the NBS-LRR structure, and also considering the conservation of LB signaling pathways between potato and tomato, candidate genes for LB resistance can be rather easily traced down. For example, the ~1.5 Mbp region of chromosome 10, to which Ph-2 has been mapped, contains only two of the 155 genes annotated as 'similar to resistance-like' (Mueller et al. 2005) or containing the aforementioned NBS-LRR domains (Moreau et al. 1998; Vleeshouwers et al. 2011) (Figure 13.2A). On the other hand, ambiguous reports exist regarding the *Ph-3* in tomato: Within the originally mapped *Ph-3* region of ~2.5 Mbp, two NBS-LRR 'hotspots' exist, and different groups have announced discovering the *Ph-3*-related molecular markers (Chunwongse et al. 2002; Zhu et al. 2006; Park et al. 2010; Foolad and Panthee 2012) (Figure 13.2B). Comprehensive comparison of potato and tomato genetic maps containing information on the location of LB-resistance traits may guide future searches for the tomato *R*-genes, previously mapped to corresponding chromosomal regions (Foolad 2007; Nowicki et al. 2012). Further information on potato LB-resistance genes is reported in Chapter 12.

Future Prospects

More than 165 years ago, LB made its mark as a destructive disease of potato. Although control of the disease was eventually attained in the 1950s and managed relatively well through the use of fungicides and semi-resistant cultivars (Fry 2008; Vleeshouwers et al. 2011), *P. infestans* remains one of the most devastating plant pathogens of all time. In potato, the identification, characterization, and introgression of new LB-resistance genes and QTLs within the past two decades have significantly contributed to the effective control of the disease. Somewhat similar but to a lesser extent, efforts have been made in tomato and some progress has been made. For example, pyramiding of the known major resistance genes *Ph-2* and *Ph-3* and recent identification of novel resistance resources and genes has increased the prospect for development of tomatoes with strong resistance against LB. Nevertheless, because of the risk of the pathogen breaching the resistance conferred by the available resistance genes, future efforts are imperative to identify new and more desirable sources of resistance and new resistance genes and QTLs. With the availability of genome sequencing data and the advent of modern high-throughput genomic methods exemplified by potato LB effectoromics, identification, cloning, characterization, and deployment of yet undiscovered tomato LB-resistance genes will not be unexpected.

Genomic Resources in Tomato

The recently sequenced potato genome of ~844 Mbp (~39,000 genes, mostly located within the ~570 Mbp of euchromatin) (Mueller et al. 2009; Xu et al. 2011) and the tomato genome of ~950 Mbp (~40,000 genes, >90% of which lie within the ~220 Mbp of euchromatin) (Mueller et al. 2005; Wang et al. 2006; http://solgenomics.net/organism/Solanum_lycopersicum/genome) significantly contribute to a larger research initiative known as the International Solanaceae Genome Project (SOL): Systems Approach to Diversity and Adaptation (http://www.sgn.cornell.edu/solanaceae-project/index.pl). The SOL Genomics Network

(SGN) database contains biological data for species in the Solanaceae family and their close relatives. Data ranges from chromosomes, maps, markers, ESTs, microarray data, and genes, to phenotypes and accessions. SGN hosts a pre-release of the tomato (*S. lycopersicum* cv Heinz 1706) reference genome sequence. It is also an open source software project, continuously developing and improving a complex system for storing, integrating, and analyzing data. The SGN curation model is community-driven, allowing researchers to add and edit up-to-date information using simple Web tools (Bombarely et al. 2011). This database facilitates a systems approach, investigating the basis of adaptation and phenotypic diversity in the Solanaceae family and other species in the Asterid clade such as coffee (*Coffea arabica*), Rubiaceae, and beyond (Mueller et al. 2005). Such a rich and broad repertoire of bio-informatic tools is an invaluable resource for both fundamental and applied science scholars. In combination with the current cutting-edge high-throughput plethora of data, SGN provides tremendous potential for research progress.

An auxilliary resource to forward- and reverse-genetics approaches in tomato studies constitutes the tomato mutant database, distributing Micro-Tom mutant collections (Saito et al. 2011). Their freely accessible database, TOMATOMA, contains 1,048 individual ethylmethane sulfonate (EMS) tomato mutant lines classified into 15 major categories and 48 subcategories, with a total of 1,819 phenotypic categories. Of these mutants, 549 were pleiotropic, whereas 499 were non-pleiotropic. A combination of the two approaches – genome comparison studies and use of forward/reverse genetic screens – may significantly enhance the tomato LB *R*-gene studies, especially when employing the modern high-throughput molecular methods such as transcriptomics and second-generation sequencing (e.g., microarray analyses). By using these technologies, researchers will be able to accrue a more complex and comprehensive view of the resistance mechanisms by which tomato defends against LB. This field of research can be strengthened even further by the input of the pathogen-related genomic resources.

Genomic Resources in *P. infestans*

Tremendous progress has been made recently in the pathogen genomics area through the availability of the second-generation sequencing of *Phytophthora* genomes. With more than 240 Mbp encoding a mere 18,155 genes in *P. infestans* (Haas et al. 2009; Raffaele et al. 2010b), its genome exhibits extremely discontinuous distribution of gene density. More than 1,400 putative disease-effector genes localize to the expanded, repeat-rich, and gene-sparse regions of the genome, which constitute ~74% of the genome size. In contrast, the housekeeping 'core ortholog' genes occupy the repeat-poor and gene-dense regions of the genome (Haas et al. 2009; Raffaele et al. 2010b). Additionally, both seem to have evolved at different paces, with the gene-sparse regions experiencing accelerated rates of evolution. Distribution patterns of the pathogen genes, which are induced during preinfection and infection stages, indicate enrichment in genes located in these gene-sparse regions (Avrova et al. 2003; Raffaele et al. 2010a; Raffaele et al. 2010b). In marked contrast, the slowly evolving gene-dense regions are enriched in genes induced in sporangia (Haas et al. 2009; Raffaele et al. 2010a; Raffaele et al. 2010b). This distinct genomic environment is thought to contribute to *P. infestans*' evolutionary potential by promoting genome plasticity, thus enhancing genetic variation of effector genes leading to host adaptation (Raffaele et al. 2010a; Raffaele et al. 2010b).

Pathogen-related genomic resources are less abundant than those for host plants, although the modern high-throughput analyses contribute a wealth of data to be utilized. An example of such input is the *Phytophthora infestans* Database, spanning the Broad Institute

laboratories from several countries, in the collective effort aiming at multiple coverage of the pathogen genome. Chromosomal and mitochondrial (haplotypes Ia, IIa, and IIb) genome sequences are available for download/analysis at http://www.broadinstitute.org/annotation/genome/phytophthora_infestans/MultiHome.html (*Phytophthora infestans* Sequencing Project, Broad Institute of Harvard and MIT). Use of such resources extends the LB-related research and contributes to fundamental research as well, in terms of comparative genomics, genetics, gene/protein annotation, plant resistance investigations, and many others.

An example of speeding-up the race with the highly variable *P. infestans* is the recent strategy for effectoromics, that is, cloning the genes, whose products display the characteristics typical for other known *P. infestans Avr* genes. Thus far, all cloned effector genes of this pathogen encode one class of proteins: The modular secreted proteins with a RXLR motif for translocation into the host cell, followed by diverse, rapidly evolving C-terminal effector domains (Vleeshouwers et al. 2011). Effectoromics, a high-throughput functional genomics approach that uses effectors to probe plant germplasm for specific recognition by R-proteins, has recently emerged as a powerful tool for identification of *Avr* and *R* genes (Jia et al. 2010; Vleeshouwers et al. 2011). The availability of genome sequence resources for *P. infestans* has enabled the generation of effector libraries cloned in vectors designed for *in planta* expression (Haas et al. 2009). Effectors are transiently expressed in *Solanum* germplasm by agro-infiltration with *Agrobacterium tumefaciens* and/or a virus vector such as Potato virus X (PVX), and plants are monitored for the occurrence of macroscopic cell death responses to the individual effectors (Vleeshouwers et al. 2011). Effectors triggering cell death represent candidate *Avr* genes. These are subsequently validated for *Avr* activity by genetic analyses. Co-segregation of the cell death response to the effectors correlates with HR-based resistance against *P. infestans* isolates in genetic populations. If the matching *R*-gene has been cloned, additional verification of R-AVR pairs can be obtained by co-expression of *R*-gene and *Avr*-gene candidates in leaves of tester plants such as *Nicotiana benthamiana* (Jia et al. 2010; Vleeshouwers et al. 2011). Similar studies with tomato are yet to be undertaken. Nevertheless, as exemplified by potato, the potential exists for the R-AVR pairs discovery of tomato proteins interacting with the pathogen AVR protein being over-expressed, for instance by means of fixation, immunoprecipitation, and subsequent analyses. Such a strategy should considerably enhance the tomato-related modern LB-resistance studies.

Conclusions

The most sustainable strategy for managing tomato LB is to deploy an integrated system including cultural practices, fungicide application, and the use of cultivars with broad-spectrum genetic resistance against LB. Unfortunately, traditional disease resistance breeding approaches can be slow and may take little advantage of the emerging knowledge of pathogenicity mechanisms. Tomato breeders now have access to at least a few LB-resistance genes of tomato origin with associated genetic markers, which can be introduced into breeding lines and commercial cultivars via marker-assisted breeding. Accumulating evidence suggests that sustainable disease resistance constitutes successful pyramiding of multiple resistance genes within a single cultivar. In tomato, the combination of *Ph-2* and *Ph-3* has been shown to confer LB resistance superior to the resistance conferred by either of the two genes alone. This superiority has been observed and confirmed by several tomato researchers. It is noteworthy, that a possible breach in the pyramided *Ph-2* and *Ph-3* was reported by R. Gardner (North Carolina State) in tomato plantings in North Carolina under highly LB-conducive conditions at the end of the season

and only in the presence of other lines with single gene resistance in the same plantings (Scott and Gardner 2007). In contrast, under severe LB infestation in multiple locations in Pennsylvania, we have observed that the resistance conferred by the combination of *Ph-2* and *Ph-3* held very well. Nonetheless, there exists a possibility for breakdown of the resistance conferred by the combination of *Ph-2* and *Ph-3*, and thus the continued search for identification and subsequent introduction of additional LB-resistance genes in tomato remains of crucial importance. Fortunately, research in this direction is underway, which will provide opportunities for pyramiding multiple resistance genes and developing tomatoes with stronger and more durable resistance against LB. This is a strategy that only will gain more significance, since predicting the extent to which an *R*-gene will continue providing resistance against a highly-variable pathogen remains problematic. Another promising research area is to combine single resistance genes with QTLs that may confer moderate but more broad-spectrum field resistance against tomato LB. Similar to work in potato, considerable efforts have been made in tomato to identify QTLs for LB resistance in different wild species, and the findings may prove to be useful for development of tomatoes with more durable resistance against LB. Furthermore, as exemplified by the fast progress in potato research and conservation of LB-signaling pathways between potato and tomato, detailed knowledge of the pathogen effectors, in combination with high-throughput genomics, may significantly contribute to identification and deployment of tomato LB-resistance sources.

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